

MEDIATEK

MT8676 Yocto AEE User Manual

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Version History

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1 AEE

1.1 Purpose

This document provides the guidelines for using the AEE feature to debug exceptions on devices.

1.2 Scope

This document is applicable to MediaTek Yocto system.

1.3 Definitions and Abbreviations

1.3.1 Definition

For the purposes of the present document, the following terms and definitions are applied.

1.3.1.1 AEE

Advanced Exception Engine (AEE) is an exception catching and debugging information generation mechanism. When an exception happens, the device will collect debug information and package into DB (database) files.

1.3.2 Abbreviation

Please note the abbreviations and their explanations are provided in [Table 1-1](#). They are used in many fundamental definitions and explanations in this document and are specific to the information that this document contains.

Table 1-1. Abbreviation

Abbreviation	Description
AEE	Advanced Exception Engine
EE	External Exception
HW Reboot	Hardware Reboot Exception
HWT	Hardware Watchdog Timeout
KE	Kernel Driver Exception
NE	Native Process Exception

1.4 Architecture Overview

1.4.1 DB Type Introduction

AEE generates different types of DB as shown in [Table 1-2](#).

Table 1-2. DB type information

Layer	Type	Level	When DB is Generated?
Native	NE	Exception	When userspace processes receive an exception signal (SIGILL/SIGABRT/SIGBUS/SIGFPE/SIGSEGV)
	System API dump		When userspace processes call AEE-provided interface
Kernel	KE	Fatal	When kernel panic occurs
	HWT		When kernel watchdog timeout occurs
	HW Reboot		When watchdog timeout occurs and the device hangs for a long time
	Kernel API dump	Exception	When kernel driver calls AEE-provided interface
External	EE	Exception	When external IC exception occurs

1.4.2 DB Type Introduction

After exception happens, AEE DB will be stored to /data/aee_exp. Those *.dbg files are DB files.

e.g.

--db.00.SystemAPI

--db.00.SystemAPI.dbg : This is a DB file which has the .dbg suffix.

--ZZ_INTERNAL : This file simply describes the exception.

--db.01.NE

--db.01.NE.dbg

--ZZ_INTERNAL

1.5 Architecture Overview

All AEE configurations are in conf file: meta/meta-mediatek-<platform>/conf/machine/<project>.conf.

1.5.1 How to Enable AEE

AEE feature will be disabled by default on customer side.

If customer needs to enable AEE, please add `MTK_LOG_CUSTOMER_SUPPORT = "yes"` to conf file.

1.5.2 Set AEE Mode

AEE mode is set to 4 by default on customer side, which only generates fatal DB. If customers want to generate normal DB and fatal DB, they need to set `setprop persist.vendor.mtk.aee.mode 3`.

If customers want to specify AEE mode during build time, please add `MTK_AEE_MODE="3"` to conf file.

1.5.3 Set Max DB Count

According to DB types described in [Table 1-2](#), the DB level includes Fatal and Normal.

The default max count of fatal DB is 8 and the default max count of Normal DB is 8.

Users can modify `MTK_AEE_FATAL_DB_CNT/MTK_AEE_DB_CNT` to change the max count.

e.g.

```
MTK_AEE_DB_CNT = "4"
```

```
MTK_AEE_FATAL_DB_CNT = "4"
```

1.6 How to Use DB

1.6.1 Pull DB

Pull platform DB to PC: `adb pull /data/aee_exp`

1.6.2 Download GAT Tool

Please search GAT on MOL to download the latest version, as shown in [Figure 1-1](#).

Filter Result							 Download Filter Result
Name	Version	Release Date ↓	Category	Product Line	Tool Type	Download	Release Note
GAT	v4.2244.1	2022-12-02	Official	ALPS	Debugging Tools	 binary	 Release Note

Figure 1-1. Download GAT tool from MediatekOnline

1.6.3 Extract DB

Open MediatekDBViewer in GAT tool. If your PC is running in a Windows operating system, please execute:

```
GAT(Official)_ALPS\GAT_exe_v4.2034.3\gat-win32-x86_64-4.2034.3.c\gat-win32-x86_64-4\tools\MediatekDBViewer.bat.
```

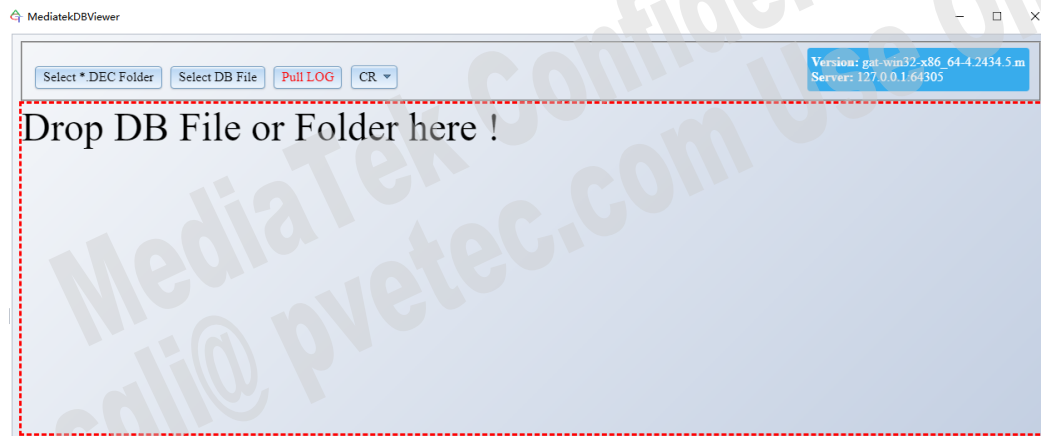


Figure 1-2. MediatekDBViewer tool UI

Click “Select DB File” button in [Figure 1-2](#), then select the DB file (*.dbg) you want to analyze and click “start” to begin extraction. All the extracted files will be listed on the left side of the UI.

The basic file is __exp_main.txt which describes the basic information of the exception.

1.6.4 Get Symbols

You may need to get debug symbols before analyzing DB. You can use `unpack-symbols.sh` to get symbols:

1. Find script: `meta/meta-mediatek/recipes-devtools/unpack-symbols-tool/files/unpack-symbols.sh`
2. Copy the `unpack-symbols.sh` to the path: `build/tmp/deploy/images/<project>/`
3. Run script: `bash unpack_symbols.sh`
4. Symbol folders will be generated in the new path, which includes all the symbols.

1.6.5 Analyze DB

1. For NE/KE/HWT issue:

Method 1:

Click “Set Symbols path” and “Analyze” (as shown in [Figure 1-3](#)), and then `out.json` file will show the detailed trace.

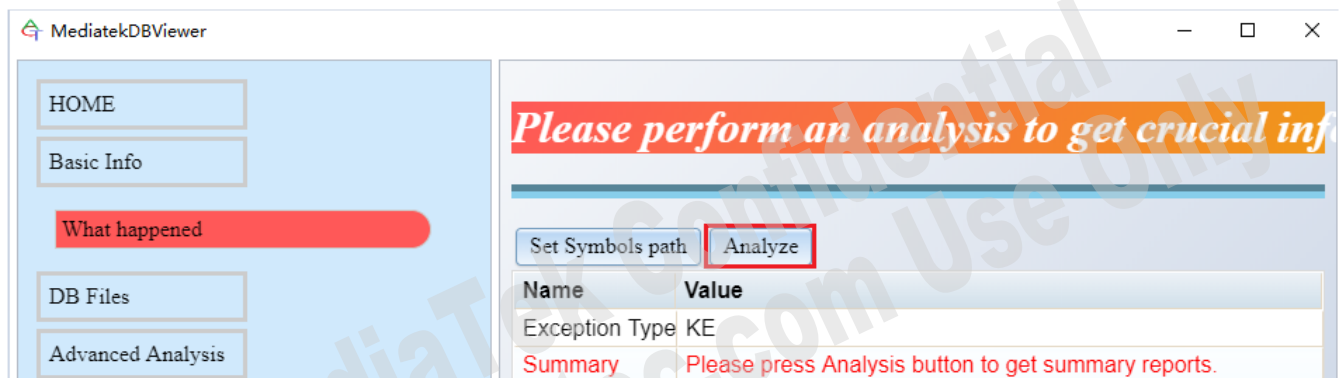


Figure 1-3. Analyze DB

Method 2:

Click “Set Symbols path” and “Launch GDB” (as shown in [Figure 1-4](#)), GDB window will show up, then you can get backtrace by using bt command in GDB.

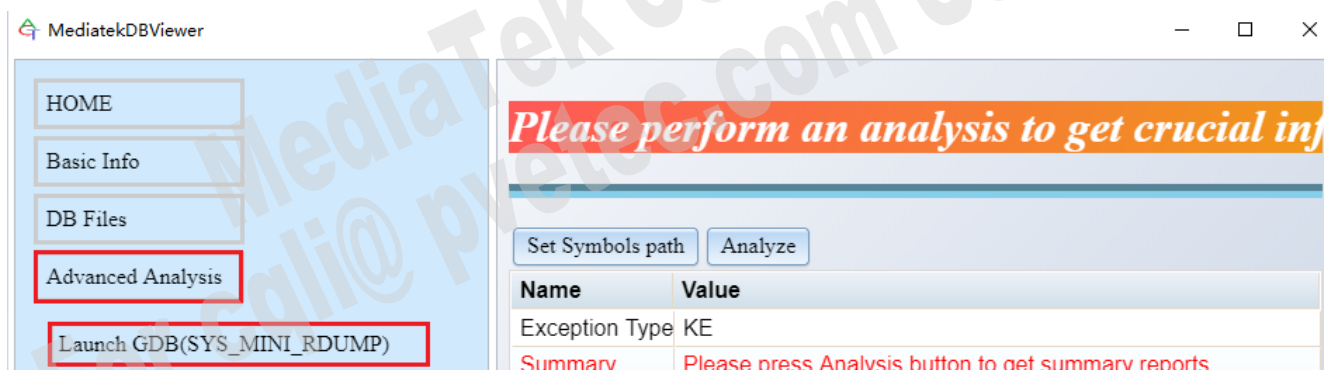


Figure 1-4. Launch GDB

2. For HW_Reboot issue:

HW_Reboot is a special issue: it cannot get backtrace, so you can only check some other debug files in DB (SYS_LAST_CPU_BUS/DFD).

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